Data Mining for Business Analytics
MSBA 511

Combining Methods: Ensembles

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Ensembles

Ensemble learning is a technique that combines the predictions of multiple individual models with the goal to create a single, stronger predictive model.

Key idea: No single model is perfect in all situations.

- Combining models can potentially correct for individual model assumptions or limitations (**Reduce bias**).
- Aggregating predictions can help smooth out errors from highly sensitive models (**Reduce variance**).
- Multiple models can increase robustness and provide stability across different datasets or scenarios.
- Different algorithms can handle different aspects of the data.

Common Ensemble Techniques

Bagging

Build multiple models on multiple random samples.

- The powerful Random Forest algorithm is one of the most popular examples of bagging.

Boosting

Sequentially train models to fix previous errors. There are a variety of powerful boosting algorithms:

1. AdaBoost (Adaptive Boosting)
2. Gradient Boosting including XGBoost and LightGBM
3. CatBoost

Voting/Stacking

Combine predictions from multiple models (voting or stacking using a meta-model approach).

Bagging

The “multiplier” effect in bagging comes from multiple bootstrap samples, rather than multiple methods.

Bootstrapping is a sampling technique where multiple datasets are created by randomly sampling **with replacement** from the original training data.

1. Start with the full training dataset of n records.
2. Randomly sample n records with replacement to create the new bootstrap sample.
3. Repeat this process many times and train a separate model on each sample.
4. Aggregate the predictions either by majority vote or average probabilities and a cutoff.

Bagging is a good technique to reduce variance since each model sees slightly different data and the aggregate of the predictions can stabilize random errors.

Boosting

Typically builds a series of **weak learners** (often shallow decision trees) where each new model focuses on improving the mistakes made by the previous models.

1. Start with a weak model on the full training data.
2. Resample records with highest weights to the records the model got wrong.
3. Build a second model that focuses on the difficult records from step 2.
4. Combine the predictions with the first model.
5. Repeat steps 2-3 for a set number of rounds or desired performance is met.

Boosting is a good technique to reduce bias since the sequentially trained models learns and gradually corrects the errors that a single simple model would make.

Voting vs. Stacking

A **Voting Classifier** is a simple combination of multiple model predictions. A **Stacking Classifier** trains a separate model (meta-learner) to combine the predictions from the base models.

Method	How it Works	Example
Hard Voting	Majority rule — each model votes a class label	2 out of 3 models predict attrition → Final prediction = attrition
Soft Voting	Average predicted probabilities across models	Average churn probability ≥ 0.5 → Final prediction = churn
Meta-Learner (Stacking)	Learns *how* to weight or combine models	Logistic Regression trained on outputs from Decision Tree, KNN, and NB

Voting Example

Accuracy:	94.90%		93.50%		88.60%		94.55%	94.25%	
	Logit Reg		**KNN**		**NB**		**Hard Voting**	**Soft Voting**	
Actual	Prob	Pred	Prob	Pred	Prob	Pred	Pred	Avg Prob	Pred
0	0.000445	0	0.0	0	0.006957	0	0	0.002467	0
0	0.020690	0	0.0	0	0.004877	0	0	0.008522	0
0	0.001346	0	0.0	0	0.007542	0	0	0.002963	0
0	0.042365	0	0.0	0	0.011038	0	0	0.017801	0
1	0.599745	1	0.0	0	0.703408	1	1	0.434384	0
1	0.508189	1	0.0	0	0.912237	1	1	0.473475	0
0	0.471148	0	0.33	0	0.742398	1	0	0.515626	1
0	0.447114	0	0.33	0	0.866027	1	0	0.548825	1
1	0.196076	0	0.0	0	0.219688	0	0	0.138588	0
1	0.992886	1	1.0	1	0.825372	1	1	0.939419	1

Use caution when averaging model probabilities since some algorithms such as Naïve Bayes produce bias probabilities.

Why not always use an Ensemble?

Simply combining multiple models does *not* guarantee better results. In some cases, voting/stacking may offer little to no improvement — or even hurt performance.

Lack of Model Diversity
Ensembles works best when the base models are diverse (i.e., different types of errors). If the model predictions are highly correlated, combining them will just make the same mistakes.

Strong Single Model
If one of your base models already performs very well on its own, adding weaker models can introduce noise and degrade performance.

Junk In ➡ Junk Out
Ensembles will not magically fix poor data quality, noisy data, and/or underperforming base models.